Stopping time and force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how a longer stop can reduce average force.

SUCCESS CRITERIA - I can...

1. Keep momentum change fixed in a fair comparison.
2. Compare force using impulse divided by time.
3. State why cushioning does not remove momentum change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg cart moving at 5 m/s stops in 2 s. Find momentum-change magnitude and average stopping-force magnitude.

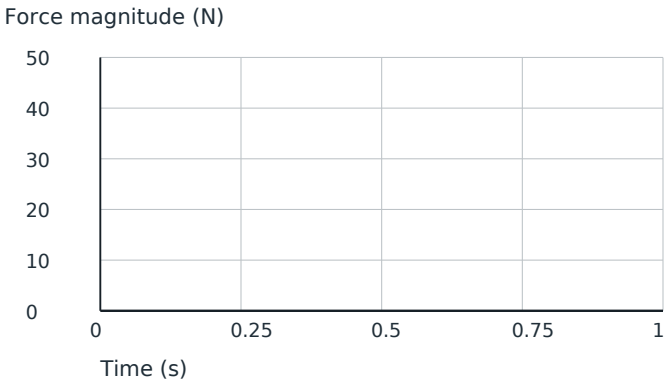
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $|F_{avg}| = |\Delta p|/\Delta t$.
For the same momentum change, more stopping time lowers average force magnitude. Bars mean magnitude.

Worked reasoning: A stopping impulse has magnitude 8 N·s. Over 0.2 s, average force is 40 N; over 0.8 s, it is 10 N. Longer stopping time lowers the average force for the same momentum change.

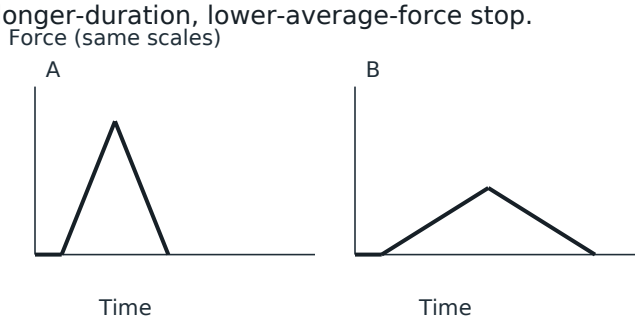
Your turn: Draw both force-time rectangles with equal areas.



Marzano check - circle: ① ② ③

03 Read and interpret the model

The two ideal force pulses have equal area. Identify the longer-duration, lower-average-force stop.



Marzano check - circle: ① ② ③

04 Apply the physics

A delivery crate loses 20 kg·m/s of momentum as it stops. Packing A stops it in 1 s; packing B stops it in 4 s. Complete the comparison table and find the average stopping-force magnitude for each packing.

Stop	Time (s)	Impulse (N·s)	Average force (N)
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

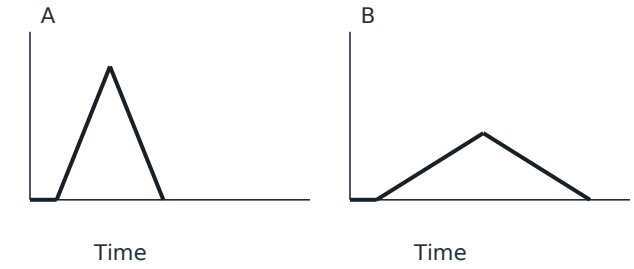
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05 Explain your reasoning

A student says a cushion makes momentum disappear. Explain where the momentum change comes from and what cushioning changes.

Force (same scales)



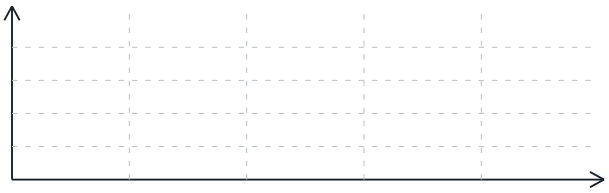
Same scales: compare impulse and peak force.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Downward velocity changes from 4 to 10 m/s in 0.6 s. Find acceleration and name one possible measurement uncertainty.
Sketch the motion and label a positive direction.

Velocity (m/s)



Marzano check - circle: ① ② ③

07 Exit check

The same object stops from the same speed in twice as much time. Compare average force sizes.

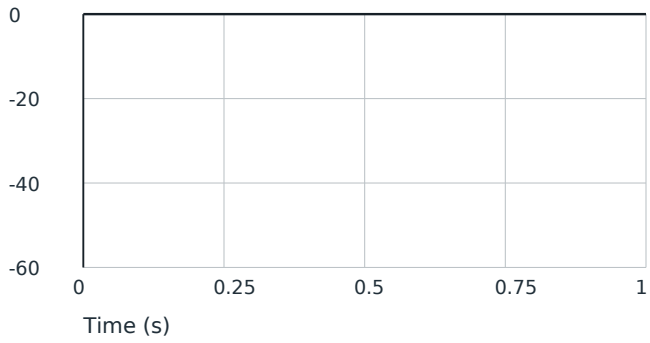
Trial	Duration	Average force
Original	Δt	F
New	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A cushion doubles stopping time without changing incoming or final velocity. Compare impulse and average force; state what cannot be concluded about the exact peak force.

Stopping force (N)



Marzano check - circle: ① ② ③